PORTFOLIO RISK ENGINE

Portfolio Risk Management Report

Returns | Volatility | Sharpe | VaR | Expected Shortfall | Monte Carlo | Stress Testing

1. Executive Summary

Data source: Yahoo Finance via yfinance. Period: 2007-04-10 au 2026-10-14. Results cover an equally weighted portfolio of 6 assets (AAPL MSFT NVDA TLT IEF GLD) with a portfolio value of 1 000 000 EUR.

17.4%	15.1%	0.95	-37.0%
Annualized Return	Annualized Volatility	Sharpe Ratio	Maximum Drawdown

Portfolio and Assumptions

Ticker	Asset Class	Weight	Notional Value
AAPL	Actions	16.7%	166 667 EUR
MSFT	Actions	16.7%	166 667 EUR
NVDA	Actions	16.7%	166 667 EUR
TLT	Obligations US	16.7%	166 667 EUR
IEF	Obligations US	16.7%	166 667 EUR
GLD	Or	16.7%	166 667 EUR
Total		100.0%	1 000 000 EUR

Risk-free rate used for the Sharpe ratio: 3.0%. Monte Carlo seed: 42. Number of simulations: 20 000. Backtest estimation window: 500 days.

2. Performance, Volatility and Diversification

Annualized returns use the mean daily log return multiplied by 252; volatility is the standard deviation of daily simple returns multiplied by the square root of 252.

Ticker	Asset Class	Annualized Return	Annualized Vol.	Sharpe	Max Drawdown
AAPL	Actions	24.6%	31.2%	0.69	-60.9%
MSFT	Actions	16.8%	28.2%	0.49	-57.9%
NVDA	Actions	31.7%	48.9%	0.59	-85.1%
TLT	Obligations US	2.6%	15.1%	-0.03	-48.4%
IEF	Obligations US	3.0%	6.9%	-0.00	-23.9%
GLD	Or	8.8%	18.2%	0.32	-45.6%

Daily Return Correlation

ρ	AAPL	MSFT	NVDA	TLT	IEF	GLD
AAPL	1.00	0.55	0.48	-0.19	-0.19	0.01
MSFT	0.55	1.00	0.53	-0.20	-0.19	0.03
NVDA	0.48	0.53	1.00	-0.17	-0.18	0.05
TLT	-0.19	-0.20	-0.17	1.00	0.91	0.17
IEF	-0.19	-0.19	-0.18	0.91	1.00	0.22
GLD	0.01	0.03	0.05	0.17	0.22	1.00

Most volatile asset: NVDA (48.9%). Average off-diagonal correlation over the full period: 0.12; during the Covid-19 episode: 0.10 (short subperiod).

3. VaR and Expected Shortfall

The measures below are expressed in euros and as positive losses when the loss quantile is positive. VaR is not a maximum loss; ES is the average loss conditional on exceeding the VaR threshold.

Method	Horizon	VaR 95%	VaR 99%	ES 95%	ES 99%
Historique	1 j	14 347 EUR	25 176 EUR	21 500 EUR	33 147 EUR
Paramétrique (normal)	1 j	14 931 EUR	21 423 EUR	18 912 EUR	24 651 EUR
Monte Carlo	1 j	14 918 EUR	20 809 EUR	18 550 EUR	23 603 EUR
Historique	10 j	41 232 EUR	71 011 EUR	59 495 EUR	88 894 EUR
Paramétrique (normal)	10 j	42 174 EUR	62 703 EUR	54 761 EUR	72 911 EUR
Monte Carlo	10 j	41 895 EUR	60 499 EUR	53 485 EUR	69 321 EUR

Model Interpretation and Differences

Historical. Empirical quantile of past losses; it depends on the estimation window and makes no parametric distribution assumption. Normal parametric. Uses the estimated mean and standard deviation under a normality assumption. Monte Carlo. Simulates multivariate normal log returns and reads the resulting simulated loss distribution; results depend on the parameters, distribution and random seed.

For the 10-day horizon, historical losses use compounded returns over rolling windows. The parametric and Monte Carlo methods assume iid daily returns in this implementation; the square-root-of-time approximation is a simplification.

Marginal Risk Contribution

Ticker	Weight	Volatility Risk Contribution
NVDA	16.7%	45.0%
AAPL	16.7%	25.0%
MSFT	16.7%	23.1%
GLD	16.7%	5.7%
TLT	16.7%	0.8%
IEF	16.7%	0.4%

4. Stress Tests and Tail Losses

Historical scenarios reconstruct buy-and-hold returns over the available window using current portfolio weights. Hypothetical shocks are specified per asset. Correlation and volatility stresses are summarized by the stressed one-day parametric 99% VaR. Ratios compare results with the reference portfolio's historical 99% VaR and ES.

Scenario	Period / Type	Loss	% Value	Loss / VaR99	Loss / ES99	Worst Asset / Factor
Crise financière 2008 (sept. 2008–mars 2009)	2008-09-02 - 2009-03-31	83 074 EUR	8.3%	3.30x	2.51x	AAPL
Covid-19 (19 fév.–23 mars 2020)	2020-02-19 - 2020-03-23	122 403 EUR	12.2%	4.86x	3.69x	NVDA
Choc inflation/taux (janv.–oct. 2022)	2022-01-03 - 2022-10-12	282 741 EUR	28.3%	11.23x	8.53x	NVDA
Hypothetical shocks	Per-asset shocks	108 333 EUR	10.8%	4.30x	3.27x	NVDA
Correlation ρ=0.90	Stressed parametric VaR 99%	34 090 EUR	3.4%	1.35x	1.03x	Correlations
Volatility x2	Stressed parametric VaR 99%	43 583 EUR	4.4%	1.73x	1.31x	Overall volatility
Reverse stress -20%	Market shock -30.6%	200 000 EUR	20.0%	7.94x	6.03x	NVDA

Interpreting the Stress Scenarios

A stress can produce a loss greater than historical VaR and ES because it imposes simultaneous moves, higher correlations or greater volatility than observed in the sample. Reverse stress estimates the approximate market shock that would reach the target loss. Ticker betas are fixed illustrative proxies, not regression estimates.

The hypothetical scenario, reverse-stress betas, and correlation/volatility stresses are conditional experiments; they do not indicate the probability of occurrence.

5. Backtesting and Project Questions

Rolling Historical 99% VaR Backtest

Statistic	Result
Out-of-sample period	2025-10-16 to 2026-10-14
Observations de test	250
Exceptions / Rate	1 / 250 (0.4%)
Expected exceptions at 99%	2.50
Kupiec LR statistic	1.176
p-value de Kupiec	0.2781
Decision at 5% level	Coverage not rejected
Traffic light	Vert (0–4 exceptions)

Summary Answers

1. Most volatile asset: NVDA, with annualized volatility of 48.9%.

2. Backtest: 1 exception(s), traffic light Vert (0–4 exceptions); Kupiec p-value 0.278. A p-value above 5% means coverage is not rejected; it does not prove the model is correct.

3. Correlations: average of 0.12 over the full period versus 0.10 during the Covid-19 window, if available; correlations can rise during crises.

4. Historical 99% VaR 25 176 EUR versus parametric VaR 21 423 EUR: empirical quantile versus normality assumption.

5. Monte Carlo 99% VaR 20 809 EUR: simulation from a fitted multivariate model, which may differ from the empirical historical tail.

6. Historical 99% ES 33 147 EUR: measures the average severity of losses beyond VaR.

7. Equal-weight concentration: HHI 0.167, or 6.0 effective positions. Greater concentration reduces potential diversification benefits.

8. Largest loss among the calculated scenarios: Choc inflation/taux (janv.–oct. 2022) (282 741 EUR), main driver: NVDA.

Conclusion

VaR provides a loss threshold, Expected Shortfall describes tail-loss severity, backtesting compares forecasts with observed exceptions, and stress tests explore situations that may not appear in the historical window. Results depend on the data, portfolio weights and model assumptions.